

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper A)

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

The following restrictions may be implemented:

- a) restrictions on the types of assets that a provider can invest in
- b) restrictions on the amount of any particular type of asset that can be taken into account for the purpose of demonstrating solvency
- c) a requirement to match assets and liabilities by currency
- d) restrictions on the maximum exposure to a single counterparty
- e) custodianship of assets
- f) a requirement to hold a certain proportion of total assets in a particular class – for example government stock
- g) a requirement to hold a mismatching reserve
- h) a limit on the extent to which mismatching is allowed at all.

(1/2 mark each max 4)

[4 Marks]

Solution 2:

Pros

- Physical gold will have security issues. Will require bank locker etc. to keep it. (1/2)
- Further, there will be cost involved in ensuring security of gold. Thus, investing on bond papers of gold will be more secure (1/2)
- Moreover, the bond provides coupons or returns annually apart from potential capital gains unlike physical gold which does not provide any annual regular returns (1)
- There is sovereign guarantee, so no risk of default and upside is fully ensured as redemption price moves in line with price of gold (1/2)
- A retail investor who requires gold at a later date, gold bond would be a better choice (1/2)

Cons

- **Illiquidity.** Lock-in period of 8 years. There is no early redemption or extension of investment period. There appears to be no secondary market as well. Physical gold is very marketable (1)
- The gold price may have significant short-term fluctuations and may fall on day of redemption due to market or other external factors. Given mandatory redemption on a fixed date, this will pose a challenge for low potential returns compared to physical gold where the investor can wait for market of gold to stabilize before selling the physical gold (1)
- Any lag in gold index or index getting changed over 8 year period might impact return (1/2)
- Any different tax treatment (favourable or unfavourable) compared to physical gold (1/2)
 - Sentimental value attached to gold /ornaments which may not be fulfilled by holding bonds (1/2)

(3 each for pros and cons total 6)

[6 Marks]

Solution 3:

Risk I

- 1) This constitutes liquidity risk (1/2)
- 2) Arises due to timing mismatch between asset and liability cash flows (1/2)
- 3) Liquidity risk arises from risks associated with projecting timing of liability cash flows accurately (1/2)
- 4) as well as from holding illiquid assets or assets becoming illiquid in adverse markets or (1/2)
- 5) The best method to determine capital to hold for this risk is to conduct a series of stress tests and hold capital based on results from such scenarios (1) for eg.,
 - a. Cash requirement arising from catastrophic events impacting portfolio eg., natcat events,
 - b. Slowdown of renewals and new business sales which leads to lower cash
 - c. Delay in recovery from reinsurers of a large claim/post a catastrophic event

Risk II

- 1) This risk constitutes an operational risk. (1/2)
- 2) Operational risks are not easily quantifiable since their frequency and amount is highly uncertain (1/2)
- 3) The best way of evaluating the capital to hold in respect of Risk II would be to use a deterministic scenario based approach. (1/2)
- 4) This involves:
 - Maintaining a risk register which contains details of all plausible adverse scenarios and developed with the input from senior management, operations team in the organisation, IT team etc (1)
 - evaluating the financial consequences of the scenario occurring. (1/2)
 - holding capital equivalent to expected financial impact from such scenarios. (1/2)

Risk III

- 1) Risk III is a credit risk. (1/2)
- 2) A lot of historical information is available on credit defaults and hence a (1/2)
- 3) Stochastic modelling approach wherein we need to allow for probability of default and loss given default with due consideration being given to rating of the non-government security; (1)
- 4) Being a small and new company stress tests may be used by assuming default in the largest holding of non-government security the company has (1)

[Max. 9 Marks]**Solution 4:**

Key risks and issues arising

A) Risk over quality of industry data

- a) Data supplied by contributors to database may be inaccurate
- b) Data may be incomplete
- c) Data quality depends on quality of all contributors' data systems
- d) As more companies contribute, more likely that one will make a mistake and invalidate the whole data set
- e) Data likely to be outdated due to time taken to collect and process

(2)

B) Risk of Relevance of data

- a) Data may not be sufficiently relevant to intended purpose such as data and experience for other pension schemes may not be comparable to this scheme, or
- b) if the company employs a different socio-economic mix than industry average, or
- c) nature of scheme benefits different from industry average or extent to which membership is voluntary or compulsory relative to average.
- d) Industry-wide data may include only a subsection of schemes and they may not be representative of the market as a whole

(2)

C) Issues in Data format

- a) Data may not be available in a desired form [1/2]for eg., it may not be available age wise, gender wise etc [1/2 for the example]
- b) Chosen data groups in the industry data might be sub-optimal for this scheme
- c) Coding used for factors by which data is split may vary between schemes

(1.5)

D) Issues Credibility of data

Volume of industry-wide data may still not be credible for some groups/ages

(1/2)

E) Risk For future projection

Past data may not reflect what will happen in future, due to

- past abnormal events
- significant random fluctuations which is more likely in a small scheme
- changes in way data was recorded
- changes in the balance of any homogenous groups
- medical changes
- social changes
- economic changes

(2.5)

F)

Issues from using data from industry without considering risks above will reduce confidence in the valuation result or may result in inadequate contribution to the scheme which could- (1/2)

Underfunding could -

1. jeopardize security of member benefits (1/2)
2. Likely increase contribution rate in future for employer (1/2)

Overfunding could lead

- to liquidity risk for employer, (1/2)
- lead to employer decreasing benefits for employees or at an extreme closure of the plan as it is considered too expensive. (1)

[Max 3 for issues on under/over funding]

[Max. 9 Marks]

Solution 5:**Mis-estimation of costs for shut down and decommissioning**

(1/2)

- 1) Estimates of the costs of shut down, decommissioning will require assumptions about:
 - a. investment returns that can be earned on the accumulated funds, if a separate fund is set aside to meet this cost (1/2)
 - b. inflation of the costs of shut down and decommissioning (1/2)
- 2) Having a detailed roll out plan in terms of when should shut down/decommissioning start for each plant and when will it end. (1/2)
- 3) Actual experience of any of these items may be different to expected, representing a risk for the Government (1/2)
- 4) A decision needs to be made as to the strength of the assumptions, and there are risks attached to an overly cautious or a too optimistic basis: (1/2)
 - a. The use of overly cautious assumptions to try to ensure sufficient funds are set aside, will lead to higher tax rate. (1/2)
Which would lead to unhappy citizens/corporations (1/2)
 - b. The use of assumptions that are too optimistic will mean that insufficient funds when the money is required. (1/2)
 - c. and the government has to either increase future taxes or reduce budgetary allocations elsewhere (1/2)
which could prove unpopular

Other risks

- 1) There are operational risks, for example:
 - a) errors in estimation of costs for eg., incorrect assumption about future inflation (1/2)
 - b) rollout plan schedule too optimistic and 10 years is too short (1/2)
 - or
 - c) 10 years too short a period of time for full switch to renewable energy sources (1/2)
- 2) Business risks-
 - a. Shut down and decommissioning means loss of coal jobs which could hurt local economy and hence government needs to ensure alternative sources of employment or spend money towards retraining/re-skilling etc. and (1)
 - b. Failure to do so means loss of political capital (1/2)
 - c. Poor tax compliance and enforcement leading to lower collection (1/2)
 - d. Alternative power sources fail to take off or take off in limited manner within the 10 year period (1/2)
 - e. Or other sources more expensive than coal plants leading to (1/2)
 - f. Double hit for citizens- tax and higher power tariffs (1/2)
 - g. Tax breaks, interest subsidies not adequate incentive for renewable energy companies to come up (1/2)
 - h. Alternative sources of power do not come up at the pace expected leading to power shortages hitting (1/2)
 - i. Economic development and growth which leads to unhappy and dissatisfied electorate (1/2)
 - j. Losing elections and being thrown out of power (1/2)
 - k. Money is invested in short term government bonds which though secure also may not be optimal way to manage surplus which ideally should be invested to match nature and liability profile of expected out go, which is mostly real and part of it could be medium to long term. (1)

[Max. 10 Marks]**Solution 6:****i) Data for the scheme in case of existing employees**

- a) Employee No
- b) Employee grade /level
- c) Date of Birth of employee
- d) Date of birth of spouse
- e) Date of joining
- f) Current salary level
- g) Retirement date/age

Additional Data in case of retirees

- h) Current level of pension drawn -50% or 100% of calculated level (If 50%, to be valued as single life and if 100%, to be taken as joint life)

(0.5 marks for each valid data point)

(Max. 4)

- ii) In order to value the various cash outgoes, a projection model will be needed to estimate for number of employees eligible for the benefits each year and their corresponding estimated salary. **(1)**

Existing employees who have completed 10 years of service**(3)**

- Need to select eligible employees (i.e. employees who have completed 10 years in service) (0.5)
- Among eligible employees, allow for probability of attrition each year in future. This attrition rate may vary based on age, no. of years of service, designation/level of the employee etc. (1)

- Among eligible employees, allow for deaths during service each year based on probability of death. The probability of death may depend on current age, type of job, gender etc. (1)
- To estimate probability of death, past death trends as well as any future mortality improvement may be allowed (0.5)

Existing employees who have completed <10 years of service (2)

- Among such employees allow for death in service, attrition in service rate which could be different to employees who have completed 10 years of service, till they complete 10 years of service at some future date (1)
- Using above, determine the employees who will become eligible every year and then project cash flows for such employees, on same basis as Existing employees who have completed 10 years of service (1)

Corresponding salary of employees (2)

- The current salary base to be taken, and then allow for future salary inflation (0.5)
- The future salary inflation may again be dependent upon level, category or type of job of employee including no. of years of service (1)
- The salary escalation rate needs to be consistent to the discount rate used for valuing liabilities (0.5)

Once we have an estimate for number of employees eligible for the benefits each year and their corresponding estimated salary, outgoes can be projected as follows

Death in service (1)

- Number of people expected to die in a year multiplied by their corresponding salary at the time of death * 3 times will be the death outgo for that year. Similar projection will give us death go for each year
- PV of this benefit is determined by discounting the death outgo each year by the discount rate

Annuities in payment post service termination (2.5)

- Single life pensioners and joint life pensioners need to be projected separately.
- Annual increase in annuity to be allowed for
- Mortality improvements in future need to be allowed for
- Projected annuity outgo for each member * probability of survival each month, will give the total outgo each month
- PV of the benefit is determined by discounting the outgo each year using the discount rate

Lump sum payments at point of termination (2.5)

- The total liability will be x% into (value of annuity), where x% is the expected proportion of people expected to take lump sum
- This proportion depends upon age of the employee during termination, health conditions of employee or spouse, financial conditions, tax incentives on taking a lump sum etc., but generally,
- There is significant anti-selection risk from those who themselves or their spouse are in bad health or member with no spouse currently
- Hence those opting for lump sum and those who do not may actually experience different mortality in future.
- Hence the present value of the annuity should be set using mortality table appropriate to those exercising this option, a heavier mortality table compared to that used in valuing annuities in payment.

Future expenses to manage the scheme (0.5)

- The expenses to manage the scheme needs to be allowed for, including inflation of such expenses

Discounting of benefits – investment return**(1)**

- The liabilities as calculated above needs to be discounted to the valuation date of the scheme.
- This discount rate will depend return on various classes the company has currently invested as well as the intention to invest in future based on its future asset allocation, allowing for default risk, reinvestment risk and liquidity risk etc.

(Max. 14)**[18 Marks]****Solution 7:****i)**

- a) The trustees main concern would be ensure that the member's benefits and reasonable expectations, in terms of any discretionary benefits (like pension increases in retirement and transfer benefits etc.) are protected and maintained. (1)
 - b) Any legal and statutory constraints to which the scheme/trust has to adhere to (1)
 - c) Trust deed and rules/documents which would determine the management of the trust and scheme (1)
- (Max 3)

ii)

The investment principle is of a defined benefit scheme would be to select investments that are appropriate to the nature, term and currency of the liabilities which would typically be long term and real. The investments should be selected so as to maximise the overall return on the assets, where overall return includes both income and capital.

(Max 2)**iii)**

To achieve this objective, ALM team needs to -

- a) Help the trustees to describe their risk appetite in terms of certain risk tolerance or limits so that it is a measurable item and can be monitored on quarterly or annual basis, for eg., (1)
- b) Trust will not accept risks that would cause its available assets below say, x% of its liabilities (1/2)
- c) Develop asset and liability projection models. This could either be deterministic or stochastic/or a combination with certain key variables like asset performance being modelled stochastically and others deterministically (1)
- d) Project any future asset liability mismatch risk and provide recommendations to the trustees based on projections on an optimal investment strategy within its risk appetite (1)
- e) Perform sensitivity analysis and stress testing of assets and liabilities under various scenarios and present the results to trustees based on likely outcome (1/2)
- f) Provide performance update of the funds and asset classes and compare its risks and rewards (1/2)
- g) Highlight risks in the fund with respect to Asset-Liability mis-matching, credit default risks or other risks such as reinvestments risk (1/2)

(Max 5)**iv)****Risks and Rewards - direct commercial property investment**

- From ALM standpoint, the liabilities are long term, real given a growing company and young employee base (1)
- Property provides real investment hence provide hedge against inflation –
 - but this will not be an exact hedge to liabilities which would typically increase with wage inflation
- Security is less compared to government security but returns higher commensurate with risk=.
 - But there is intrinsic value in the land in which property is built
 - the returns expected to be higher than index linked bonds due to inherent risk involved

- Regular income through rentals can be used to meet ongoing liabilities from deaths, transfers –
 - but can be unstable in times of economic down turn due to tenant default
 - risk of void especially during economic downturn
 - Rental income increase could also be regulated by local laws, which could bring some level of uncertainty of projections of rental income and cash flows from property investments
 - Marketability of properties is lower compared to other assets, hence need to ensure adequate liquid/marketable resources through minimum investments in other asset classes which are more marketable and liquid (1)
- Following unique features of commercial property investment also need to be allowed while valuing property for any ALM exercise-
- Valuation of property a challenge hence need expert opinions
 - Risk of depreciation of value / obsolescence is very unique to property as an asset class
 - Cost of maintaining property is much higher than other asset classes- Insurance costs and maintenance costs
 - Property is also an indivisible asset, hence cannot sell a part of the investment,

Other risks:

- There are financial and reputations risks associated with eviction of a tenant
- Needs good investment expertise otherwise risk from making poor choices
- Could lead to concentration risk and it depends on the diversification that can be achieved across geographies, types of commercial property, type of tenants etc. (1)

(1/2 each unless otherwise stated, Max 10)

[20 Marks]

Solution 8:

i)

For Y Electronics

- May want to reduce exposure of credit risk specially when business has grown and would prefer to exchange the uncertain future credit risk for a certain current amount.
- They may be in need of liquidity

(2)

For AACL

- AACL will make money out of this deal if they pay less for this block of business than the present value of the future repayments.
- They also potentially improve the diversification of their existing business and/or improve the economies of scale in their existing business to reduce overall expense.

(2)

(4)

ii)

- AACL would need to consider whether the parameters and the model is able to predict the expected experience of Y Electronics clients accurately.
- This can be done by checking historic data from Y Electronics and run actual vs. expected credit defaults analysis based on last 5 years data

The key reasons for deviation should be investigated further for:

- Is the in-house data of AACL is comparable to Y's in terms of socio-economic sections represented?

- What types of industries are represented in the experience?
- What sort credit checks/scoring was done by the other blocks of debt in the experience compared to Y Electronics- quality of credit underwriting.
- If the credit was extended more easily then the experience might be worse than that of Y Electronics.
- The same principle applies in considering how diligently outstanding debt payments are followed up by the client.
- What were the average credit amounts and repayment terms and duration compared to the Y Electronics book?
- How accurate is the data recording from the other companies compared to Y Electronics fewer/more historical errors lead to more/less reliable modelling.
- Are there any features and/or trends in the in-house data that are perhaps not expected to be present in the near future?
- These may include things like an economic recession, or periods of high interest rates and/or inflation

The model and the underlying parameters need to be suitably modified to reflect above differences before use for determining the value of Y Electronic debt.

(1/2 each, max 6)

iii)

Deterministic Model

- Parameters are fixed at outset ...
- Result of running model is a single outcome
- Potential variability is assessed by sensitivity analysis and scenario testing

(2)

iv)

Stochastic model

- At least one parameter is estimated by assigning it a probability distribution
- Model run large no of times with value of stochastic parameters randomly selected from their distribution on each run
- Outcome is range of values

(2)

v)

- The most important decision to make would be to decide which parameters should be stochastic and which should be deterministic. (1/2)
- Since here credit default is the key parameter, it is important to model this stochastically
..... a suitable model (like normal, log normal etc.) need to be finalised and calibrated. (1)
- Credit default rate is also correlated to other key parameters like interest rate and inflation rate. (1/2)
- Hence decision has to be made as to how to define the correlation between these various parameters, which (1/2)
- Can be done studying historical behaviour , both for Y Electronics, as well as other clients for AACL (1/2)
- For calibrating/parameterising the model, you should collect, group and analyse the past data from Y Electronics (1/2)
- The grouping of data could be based on age, annual income, EMI bands or other key factors which drive credit default behaviour (1)
- The grouping should be chosen in such a way that the volume of data in each cell is statistically credible and homogenous (1/2)
- Once the model is suitably parameterised for various groups, (3)

- for each group you will project a cash flows for each future period of time which is likely to be monthly or a for short period.
 - The monthly amount due is deterministic and typically known
 - Using the values generated by the credit model you can simulate the amount that the client pays [including zero in the case of a default].
 - It may be necessary to specify some correlations between repayments for example, if a client defaulted already, he may be at higher risk of defaulting again.
- You can then discount the payments(which could be different for each scenario due to correlation between the parameters) to the present and aggregate to come up with a lump sum representing the present value of future payments for that iteration of the simulation. (1)
 - You may also need to project expenses associated with managing the block and allow for future inflation in expenses, which again could vary with each scenario, due to correlation between inflation and credit default rate (1)
 - You may then simulate multiple times say 10,000 runs, to derive a distribution of the present value of future debt repayments on this block. (1/2)
 - Using the results [or summary thereof] you can choose a lump sum representing the percentile [or level of confidence] at which we are comfortable to take over this book. (1/2)
 - You can also perform sensitivity and scenario checks by tweaking parameters in the model (1/2)
- (Max 10,no marks for generic responses)
[24 Marks]
